

Radip Shrestha

Highland Heights, KY • (859) 287-5362 • radip.crestha@gmail.com

Objective

Computer Science major graduating in May 2026 with experience developing AI-driven applications, real-time computer vision systems, and backend architectures. Seeking a full-time Software Engineering role to build scalable services and deploy high-impact, intelligent software solutions.

Technical Skills

Languages: Python, C++, Java, JavaScript, PHP, C#, SQL, NoSQL

Frameworks & Tools: Flask, REST API Development, Git, Linux, Nextcloud, Bash, AWS

Databases: PostgreSQL, MongoDB

Data & Visualization: Pandas, NumPy, Matplotlib

System Administration: Ubuntu Server Setup, Networking, File Permissions, Privilege Management, Environment Inheritance

Certifications: Microsoft C# Certification (in progress)

Projects

RAG Fitness Assistant (Python, Flask, LangChain, ChromaDB, SQLite, HuggingFace (Qwen 2.5, Transformers, BitsAndBytes) and Chart.js.)

- A local RAG system that generates personalized workouts using hybrid BM25 + vector search over a custom exercise dataset.
- Automatically extracts weight and PR updates from chat and stores them in SQLite, powering a real-time progress dashboard.
- Implemented custom LLM interceptors to control state, prevent hallucinations, and enforce structured multi-day workout logic.

Real-Time Exercise Classifier & Form Analyzer (Python, MediaPipe, TensorFlow/Keras, OpenCV)

- Designed a Bi-LSTM + Temporal Attention model to recognize 7 exercise types and generate a continuous form-quality score.
- Implemented a multi-task learning pipeline using pose landmarks (33 points) for simultaneous classification and form regression.
- Achieved 90%+ accuracy with low-latency inference using a sliding-window buffer, enabling smooth, real-time feedback on consumer hardware.

Library Management System (PHP, PostgreSQL, RBAC)

- Developed a full-stack library management platform with secure authentication and role-based access for Librarian and Member workflows.
- Built modules for cataloging, issuing/returning books, user accounts, and activity logging.
- Optimized PostgreSQL queries and indexing strategies to improve catalog search performance.
- Designed a clean, responsive dashboard to streamline daily operations.

Supplement Store Location Recommendation System (Python, DBSCAN, Geospatial Analysis)

- Built a geospatial clustering model using DBSCAN to identify optimal supplement store locations based on gym density and competitor proximity.
- Processed and visualized spatial data using Pandas, NumPy, and Matplotlib.
- Generated density maps and cluster reports to support strategic business placement decisions.

Personal Cloud Server Deployment (Ubuntu, Nextcloud)

- Repurposed an old laptop into a secure, self-hosted cloud server using Nextcloud on Ubuntu.
- Configured remote access, user permissions, SSL, and automated backup routines.
- Documented the full deployment process for reproducibility and scalability.

Education

Northern Kentucky University — Bachelor of Science in Computer Science

Graduation: May 12, 2026

Minor: Computer & Information Technology

Relevant Coursework

Data Structures & Algorithms, Operating Systems, Database Systems, Software Engineering, Computer Networks, Machine Learning